

01 Transaction Analytics Berapa total volume dan nilai transaksi per hari, minggu, dan bulan? Apa tren pertumbuhannya? Tables: dim_date · dim_channel · fact_transactions	02 Customer 360 Siapa nasabah paling aktif berdasarkan frekuensi dan nilai transaksi? Bagaimana distribusi per segmen (Retail/Priority/VIP)? Tables: dim_customer · dim_account · fact_transactions
03 Branch Performance Cabang mana yang memiliki performa tertinggi berdasarkan jumlah transaksi dan total nilai transaksi per region? Tables: dim_branch · dim_date · fact_transactions	04 Channel Analysis Channel apa yang paling banyak digunakan nasabah (ATM, Mobile, Teller, Internet Banking)? Bagaimana tren migrasi ke digital? Tables: dim_channel · dim_date · fact_transactions
05 Product Performance Produk rekening mana (Tabungan/Giro/Deposito) yang menghasilkan volume transaksi dan saldo rata-rata tertinggi? Tables: dim_account · dim_date · fact_transactions	06 Risk & Fraud Detection Adakah transaksi anomali (nilai sangat besar, frekuensi tidak wajar, atau status FAILED berulang) yang perlu diwaspadai? Tables: dim_customer · dim_channel · fact_transactions

Jawaban Pertanyaan Bisnis

1. Transaction Analytics

a. Per Hari

Query:

```
SELECT
  d.full_date,
  COUNT(ft.transaction_id) AS total_transactions,
  SUM(ft.amount) AS total_amount
FROM fact_transactions ft
JOIN dim_date d
  ON ft.date_id = d.date_id
GROUP BY d.full_date
ORDER BY d.full_date;
```

Output:

#	full_date	total_transactions	total_amount
1	2023-01-01	47	193820215.00
2	2023-01-02	38	120153959.00
3	2023-01-03	49	147891191.00
4	2023-01-04	41	149278992.00
5	2023-01-05	44	156755101.00
6	2023-01-06	44	146784421.00
7	2023-01-07	47	136763304.00
8	2023-01-08	51	176989803.00
9	2023-01-09	43	117067565.00
10	2023-01-10	46	177788081.00
11	2023-01-11	55	190458895.00
12	2023-01-12	49	194182625.00
13	2023-01-13	62	163732577.00

b. Per Minggu

Query:

```

SELECT
    EXTRACT(YEAR FROM d.full_date) AS year,
    EXTRACT(WEEK FROM d.full_date) AS week_of_year,
    MIN(d.full_date) AS week_start,
    MAX(d.full_date) AS week_end,
    COUNT(ft.transaction_id) AS total_transactions,
    SUM(ft.amount) AS total_amount,
    ROUND(AVG(ft.amount),2) AS average_amount
FROM fact_transactions ft
JOIN dim_date d
    ON ft.date_id = d.date_id
GROUP BY
    EXTRACT(YEAR FROM d.full_date),
    EXTRACT(WEEK FROM d.full_date)
ORDER BY
    year,
    week_of_year;

```

Output:

#	year	week_of_year	week_start	week_end	total_transactions	total_amount	average_amount
1	2023	1	2023-01-02	2023-01-08	314	1033816771.00	3292410.10
2	2023	2	2023-01-09	2023-01-15	319	1113564155.00	3490796.72
3	2023	3	2023-01-16	2023-01-22	324	1198755401.00	3699862.35
4	2023	4	2023-01-23	2023-01-29	322	1111153526.00	3450787.35
5	2023	5	2023-01-30	2023-02-05	337	1192811969.00	3539501.39
6	2023	6	2023-02-06	2023-02-12	321	1048466979.00	3266252.27
7	2023	7	2023-02-13	2023-02-19	365	1277963314.00	3501269.35
8	2023	8	2023-02-20	2023-02-26	331	1140015150.00	3444154.55
9	2023	9	2023-02-27	2023-03-05	311	1026330610.00	3300098.42
10	2023	10	2023-03-06	2023-03-12	314	1083080436.00	3449380.75
11	2023	11	2023-03-13	2023-03-19	334	1098618769.00	3289277.75
12	2023	12	2023-03-20	2023-03-26	304	1026924799.00	3378842.10
13	2023	13	2023-03-27	2023-04-02	317	1043642749.00	3292248.42
14	2023	14	2023-04-03	2023-04-09	292	954719101.00	3269580.24
15	2023	15	2023-04-10	2023-04-16	297	1059041917.00	3535491.64
16	2023	16	2023-04-17	2023-04-23	311	1080341939.00	3473768.29
17	2023	17	2023-04-24	2023-04-30	290	1030832576.00	3554595.99
18	2023	18	2023-05-01	2023-05-07	331	1079925112.00	3262613.63
19	2023	19	2023-05-08	2023-05-14	326	1038734227.00	3186301.31
20	2023	20	2023-05-15	2023-05-21	344	1115760794.00	3243490.68
21	2023	21	2023-05-22	2023-05-28	312	974240761.00	3122566.54
22	2023	22	2023-05-29	2023-06-04	284	954104321.00	3465156.06

c. Per Bulan

Query:

```

SELECT
    d.year,
    d.month,
    d.month_name,
    COUNT(ft.transaction_id) AS total_transactions,
    SUM(ft.amount) AS total_amount
FROM fact_transactions ft

```

```

JOIN dim_date d
ON ft.date_id = d.date_id
GROUP BY
    d.year,
    d.month,
    d.month_name
ORDER BY
    d.year,
    d.month;

```

Output:

#	year	month	month_name	total_transactions	total_amount
1	2023	1	Januari	1436	5045173863.00
2	2023	2	Februari	1332	4553522157.00
3	2023	3	Maret	1394	4665495366.00
4	2023	4	April	1288	4439908176.00
5	2023	5	Mei	1418	4568271446.00
6	2023	6	Juni	1402	4602770270.00
7	2023	7	Juli	1426	4894715856.00
8	2023	8	Agustus	1406	4702867146.00
9	2023	9	September	1427	4786698152.00
10	2023	10	Oktober	1486	4820723487.00
11	2023	11	November	1330	4494241569.00
12	2023	12	Desember	1432	4742667256.00
13	2024	1	Januari	1482	4940403075.00

2. Customer 360

- Nasabah paling aktif berdasarkan frekuensi transaksi

```

Query:
SELECT
    c.customer_id,
    c.full_name,
    COUNT(ft.transaction_id) AS total_transactions,
    SUM(ft.amount) AS total_amount
FROM fact_transactions ft
JOIN dim_customers c
ON ft.customer_id = c.customer_id
GROUP BY
    c.customer_id,
    c.full_name
ORDER BY total_transactions DESC
LIMIT 10;

```

Output:

#	year	month	month_name	total_transactions	total_amount
1	2023	1	Januari	1436	5045173863.00
2	2023	2	Februari	1332	4553522157.00
3	2023	3	Maret	1394	4665495366.00
4	2023	4	April	1288	4439908176.00
5	2023	5	Mei	1418	4568271446.00
6	2023	6	Juni	1402	4602770270.00
7	2023	7	Juli	1426	4894715856.00
8	2023	8	Agustus	1406	4702867146.00
9	2023	9	September	1427	4786698152.00
10	2023	10	Oktober	1486	4820723487.00
11	2023	11	November	1330	4494241569.00
12	2023	12	Desember	1432	4742667256.00
13	2024	1	Januari	1482	4940403075.00

b. Distribusi per segmen

Query:

```
SELECT
    c.segment,
    COUNT(ft.transaction_id) AS total_transactions,
    SUM(ft.amount) AS total_amount,
    COUNT(DISTINCT c.customer_id) AS total_customer
FROM fact_transactions ft
JOIN dim_customers c
ON ft.customer_id = c.customer_id
GROUP BY c.segment
ORDER BY total_amount DESC;
```

Output:

#	segment	total_transactions	total_amount	total_customer
1	RETAIL	40544	135372068379.00	1495
2	PRIORITY	6901	22898547122.00	242
3	VIP	2555	8510563107.00	87

3. Branch Performance

a. Cabang terbaik berdasarkan volume dan nilai transaksi

Query:

```
SSELECT
    b.branch_name,
    b.region,
    COUNT(ft.transaction_id) AS total_transactions,
    SUM(ft.amount) AS total_amount
FROM fact_transactions ft
JOIN dim_branches b
ON ft.branch_id = b.branch_id
```

```
GROUP BY
    b.branch_name,
    b.region
ORDER BY total_amount DESC;
```

Output:

#	branch_name	region	total_transactions	total_Amount
1	KCU Palembang Sudirman	SUMSEL	2550	8544795997.00
2	KCP Yogyakarta Malioboro	JATENG	2585	8440608956.00
3	KCU Pekanbaru Sudirman	SUMSEL	2379	7856682861.00
4	KCP Medan Helvetia	SUMUT	2291	7720988449.00
5	KCP Jember Gajah Mada	JATIM	2275	7660916800.00
6	KCP Cirebon Kartini	JABAR	2155	7414651702.00
7	KCU Semarang Pemuda	JATENG	2202	7240734923.00
8	KCU Makassar Ahmad Yani	SULAWESI	2188	7236812342.00
9	KK Bogor Pajajaran	JABODETABEK	2103	7023163881.00
10	KCU Jakarta Selatan	JABODETABEK	2117	6942190679.00
11	KCU Surabaya Rungkut	JATIM	2052	6934641636.00
12	KCP Malang Kawi	JATIM	2043	6754946246.00
13	KCU Jakarta Utara	JABODETABEK	1995	6630400189.00
14	KCP Bandung Dago	JABAR	1921	6487802869.00
15	KCP Tangerang Selatan	JABODETABEK	1939	6475539012.00
16	KCP Depok Margonda	JABODETABEK	1951	6358789477.00
17	KCU Surabaya Tunjungan	JATIM	1910	6240162472.00
18	KCU Bandung Asia Afrika	JABAR	1862	6136300834.00
19	KCP Kuta Legian	BALI-NUSA	1744	5953376477.00
20	KCP Bekasi Jaya	JABODETABEK	1783	5792084142.00
21	KCP Solo Slamet Riyadi	JATENG	1660	5699984166.00
22	KCU Jakarta Pusat	JABODETABEK	1680	5594651049.00

b. Top Region

```
Query:
SELECT
    b.region,
    COUNT(*) AS total_transactions,
    SUM(ft.amount) AS total_amount
FROM fact_transactions ft
JOIN dim_branches b
ON ft.branch_id = b.branch_id
GROUP BY b.region
ORDER BY total_amount DESC;
```

Output:

#	region	total_transactions	total_Amount
1	JABODETABEK	13568	44826824429.00
2	JATIM	8200	27590667154.00
3	JATENG	6367	21381148045.00
4	JABAR	5938	20032755405.00
5	SUMSEL	4929	16401478058.00
6	SULAWESI	3834	12768250014.00
7	SUMUT	3731	12596876995.00
8	BALI-NUSA	3363	11183178508.00

4. Channel Analysis

a. Channel yang paling sering digunakan

Query:

```
SELECT
    ch.channel_name,
    COUNT(*) AS total_transactions,
    SUM(ft.amount) AS total_amount
FROM fact_transactions ft
JOIN dim_channels ch
ON ft.channel_id = ch.channel_id
GROUP BY ch.channel_name
ORDER BY total_transactions DESC;
```

Output:

#	channel_name	total_transactions	total_amount
1	Mobile Banking	17568	59062180200.00
2	ATM	10041	33226587542.00
3	Teller	9927	32855009219.00
4	Internet Banking	7455	24881632306.00
5	EDC / Mesin Kasir	2538	8534582436.00
6	Call Center / IVR	2471	8221186905.00

b. Tren penggunaan channel digital

Query:

```
SELECT
    d.year,
    d.month,
    ch.channel_category,
    COUNT(*) AS total_transactions,
    SUM(ft.amount) AS total_amount
FROM fact_transactions ft
JOIN dim_channels ch
ON ft.channel_id = ch.channel_id
JOIN dim_date d
ON ft.date_id = d.date_id
GROUP BY
    d.year,
    d.month,
    ch.channel_category
ORDER BY
    d.year,
    d.month;
```

Output:

#	year	month	channel_category	total_transactions	total_amount
1	2023	1	DIGITAL	780	2762501390.00
2	2023	1	PHYSICAL	656	2282672473.00
3	2023	2	PHYSICAL	689	2855237836.00
4	2023	2	DIGITAL	723	2498284321.00
5	2023	3	PHYSICAL	648	2176828519.00
6	2023	3	DIGITAL	746	2489474847.00
7	2023	4	PHYSICAL	582	2804397338.00
8	2023	4	DIGITAL	706	2435510838.00
9	2023	5	DIGITAL	799	2606303877.00
10	2023	5	PHYSICAL	619	1961968389.00
11	2023	6	PHYSICAL	648	2864599977.00
12	2023	6	DIGITAL	754	2538178293.00
13	2023	7	PHYSICAL	618	2897181564.00
14	2023	7	DIGITAL	808	2797534292.00
15	2023	8	DIGITAL	767	2477120963.00
16	2023	8	PHYSICAL	639	2225746183.00
17	2023	9	DIGITAL	783	2621711567.00
18	2023	9	PHYSICAL	644	2164986585.00
19	2023	10	PHYSICAL	684	2237382165.00
20	2023	10	DIGITAL	802	2583341322.00
21	2023	11	DIGITAL	769	2616676742.00
22	2023	11	PHYSICAL	561	1877564827.00

5. Product Performance

- Produk rekening paling banyak digunakan

Query:

```
SELECT
```

```
    a.product_name,
    a.account_type,
    COUNT(ft.transaction_id) AS total_transactions,
    SUM(ft.amount) AS total_amount,
    AVG(ft.balance_after) AS avg_balance
```

```
FROM fact_transactions ft
```

```
JOIN dim_accounts a
```

```
ON ft.account_id = a.account_id
```

```
GROUP BY
```

```
    a.product_name,
    a.account_type
```

```
ORDER BY total_amount DESC;
```

Output:

#	product_name	account_type	total_transactions	total_amount	avg_balance
1	Tabungan Bisnis	TABUNGAN	7606	25588748277.00	49543435.515908493295
2	Tabungan Online	TABUNGAN	7508	25131373383.00	49369383.593766648908
3	Tabungan Junior	TABUNGAN	7519	24766189476.00	49339299.024338342865
4	Tabungan Reguler	TABUNGAN	7123	23755541991.00	49235525.293836866409
5	Giro Valas	GIRO	3487	11684643008.00	49258695.990823057869
6	Giro Reguler	GIRO	3467	11480649129.00	49552662.488530719200
7	Giro Bisnis	GIRO	3165	10613328221.00	49125816.675197472354
8	Deposito 1 Bulan	DEPOSITO	3078	10214145267.00	48991477.140772579597
9	Deposito 6 Bulan	DEPOSITO	2581	8782182963.00	48450115.873692307299
10	Deposito 12 Bulan	DEPOSITO	2480	8343128283.00	50068476.755241935484
11	Deposito 3 Bulan	DEPOSITO	1986	6581264610.00	49368328.928946626385

b. Performa berdasarkan jenis rekening

Query:

```
SELECT
    a.account_type,
    COUNT(ft.transaction_id) AS total_transactions,
    SUM(ft.amount) AS total_amount,
    AVG(ft.balance_after) AS average_balance
FROM fact_transactions ft
JOIN dim_accounts a
ON ft.account_id = a.account_id
GROUP BY a.account_type
ORDER BY total_amount DESC;
```

Output:

#	account_type	total_transactions	total_amount	average_balance
1	TABUNGAN	29756	99241853127.00	49374228.343460142492
2	GIRO	10119	33778612358.00	49317853.890404190137
3	DEPOSITO	10125	33760713123.00	49191194.007604938272

6. Risk & Fraud Detection

a. Transaksi fraud

Query:

```
SELECT
    ft.transaction_id,
    c.full_name,
    ft.amount,
    ft.transaction_type,
    ft.status,
    ft.fraud_type,
    ft.fraud_score,
    ft.transaction_at
FROM fact_transactions ft
JOIN dim_customers c
ON ft.customer_id = c.customer_id
WHERE ft.is_fraud = TRUE
```


ORDER BY ft.fraud_score DESC;

Output:

#	transaction_id	full_name	amount	transaction_type	status	fraud_type	fraud_score	transaction_at
1	33915	Oki Wahyudi	1765922.00	PEMBAYARAN	SUCCESS	CARD_SKIMMING	0.9897	2024-08-01 17:23:22
2	9038	Indah Wahyudi	1000000.00	TARIK_TUNAI	SUCCESS	PHISHING	0.9894	2025-09-22 14:31:26
3	33400	Putra Hidayat	4025170.00	DEBIT	SUCCESS	CARD_SKIMMING	0.9870	2024-07-19 11:28:03
4	10547	Kartika Purnomo	9879787.00	TRANSFER_OUT	SUCCESS	SIM_SWAP	0.9848	2024-12-18 14:07:14
5	2400	Rina Siregar	4057641.00	TRANSFER_IN	SUCCESS	UNUSUAL_AMOUNT	0.9846	2024-09-15 20:51:21
6	20317	Sandi Prasetyo	5526975.00	TRANSFER_OUT	SUCCESS	UNUSUAL_AMOUNT	0.9845	2023-06-10 19:05:51
7	21067	Cahyo Safitri	1806525.00	DEBIT	SUCCESS	CARD_SKIMMING	0.9840	2024-08-25 14:49:12
8	9610	Ayu Wijaya	7741426.00	TRANSFER_IN	SUCCESS	UNUSUAL_AMOUNT	0.9835	2023-02-23 13:22:18
9	36936	Oki Hermawan	300000.00	SETOR_TUNAI	SUCCESS	CARD_SKIMMING	0.9827	2023-01-27 21:27:42
10	375	Tini Santoso	4083948.00	DEBIT	SUCCESS	SIM_SWAP	0.9825	2024-07-13 10:17:03
11	39022	Sandi Wahyudi	1513013.00	KREDIT	SUCCESS	CARD_SKIMMING	0.9821	2025-09-15 13:00:00
12	4798	Hana Lubis	4385400.00	DEBIT	SUCCESS	ACCOUNT_TAKEOVER	0.9817	2023-07-08 07:40:17
13	31402	Vino Harahap	9925142.00	TRANSFER_OUT	SUCCESS	SIM_SWAP	0.9817	2025-03-05 13:47:38
14	42050	Dimas Siregar	8670015.00	TRANSFER_IN	SUCCESS	CARD_SKIMMING	0.9813	2023-07-21 14:45:11
15	5105	Xander Saputra	1536189.00	KREDIT	SUCCESS	SIM_SWAP	0.9800	2023-01-27 14:12:28
16	11599	Eko Susanto	4719519.00	DEBIT	SUCCESS	SIM_SWAP	0.9798	2025-03-23 17:32:34
17	41986	Gori Nugroho	4859046.00	DEBIT	SUCCESS	CARD_SKIMMING	0.9795	2025-09-30 09:22:37
18	10571	Nita Safitri	6240428.00	PEMBAYARAN	SUCCESS	VELOCITY_ABUSE	0.9789	2024-02-25 14:37:22
19	41845	Tono Sihombing	2596857.00	TRANSFER_IN	SUCCESS	UNUSUAL_AMOUNT	0.9777	2023-05-13 00:12:22
20	31114	Liana Manurung	3370659.00	KREDIT	SUCCESS	VELOCITY_ABUSE	0.9775	2024-07-10 13:16:17
21	12290	Budi Harahap	3065809.00	DEBIT	SUCCESS	SIM_SWAP	0.9772	2024-06-21 11:56:46
22	47853	Budi Rahayu	4305544.00	DEBIT	SUCCESS	UNUSUAL_AMOUNT	0.9753	2023-05-09 13:38:07

b. Transaksi FAILED terbanyak

Query:
SELECT
 c.full_name,
 COUNT(*) AS failed_transactions
FROM fact_transactions ft
JOIN dim_customers c
ON ft.customer_id = c.customer_id
WHERE ft.status = 'FAILED'
GROUP BY
 c.full_name
HAVING COUNT(*) >= 3
ORDER BY failed_transactions DESC;

Output:

#	full_name	failed_transactions
1	Xena Manurung	29
2	Yanuar Panjaitan	26
3	Sari Nugroho	24
4	Agus Hartono	23
5	Eko Hermawan	23
6	Eko Wijaya	23
7	Zaki Setiawan	23
8	Citra Wibowo	22
9	Rizki Nasution	22
10	Maman Manurung	21
11	Lukman Santoso	21
12	Fitri Sihombing	21
13	Olga Setiawan	21
14	Rina Gunawan	21
15	Lukman Susanto	21
16	Xander Manurung	21
17	Kevin Wijaya	19
18	Ulfa Prasetyo	19
19	Yuli Anggraini	19
20	Julia Prasetyo	19
21	Dewi Saputra	19
22	Eko Hidayat	19